SIDDARTH BOGGARAPU

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Student in Bangalore. Built an 8-bit breadboard computer from 74HC logic chips, plus software that measures language models, correlates security alerts, and refuses answers it cannot support. CSET / SaTML submissions are **under review** (not accepted, not published).

RESEARCH / WORKSHOP SUBMISSIONS (PENDING)

CSET · SaTML — under review

Security experimentation (CSET) and trustworthy ML (SaTML) · decisions pending

- Submitted and under review. Not accepted or published.

VENTURES AND PROFESSIONAL EXPERIENCE

Orqis — Technical cofounder

Live product · FastAPI · libcst · MCP · Redis · human review before merge

- Owns backend for an agent-ops product: detect runaway loops, explain via MCP, open a reviewable patch/PR.
- Deterministic libcst remediations plus confidence-gated LLM assist; never silent-pushes the default branch.
- Live product with beta onboarding.

Athera (athera.digital) — Founder

AI automation and websites for small businesses · 6+ months

- Two client websites plus an in-progress AI HR build for a client; client names private.
- Ops glue across sheets, Gmail, webhooks, and live deploys.

VidhiSetu (VidhiSethu / Lexprobe) — Founder / builder

Indian-legal RAG with citation audit · closed beta

- Natural-language legal research over an Indian corpus; citation-audit layer filters unsupported claims.
- Refined with informal feedback from 2 law agencies and 3 lawyers; public architecture docs; application remains private.

Evex — Founder / lead, event company (Bangalore)

Operations and P&L

- Hosted 20+ school graduation parties across Bangalore; about INR 5 lakh profit.
- Leadership, logistics, and client delivery outside pure coding.

INTERNSHIPS

Indian Institute of Science (IISc) — Aerodynamics intern

May 2025 · 10 days

- Lab exposure to vortex-ring formation and instability regimes.
- Built vortex-tracker (OpenCV ring diameter and propagation speed) during the internship.
- Follow-on propeller and vortex-ring simulation work was self-directed, on my own time.

Vegam Solutions — Engineering intern (two roles)

Air-filtration hardware (2025 Q3) · Text-to-SQL RAG, 1 month full-time (2026 Q1, NDA)

- Hardware: end-to-end water-based air-filtration prototype under shop constraints.
- Software: NL→SQL retrieval with schema awareness and grounded refusal; public clean-room companion: text2sql-rag (Spider evaluation pending).

SELECTED PROJECTS (TECHNICAL)

Discrete-logic CPU — built 8-bit breadboard computer

74HC logic on six breadboards · ESP8266 loads the program only · not a simulated CPU

- Built the processor from 74HC chips: registers, adder, decoders, and a 555 clock, on six solderless breadboards.
- The ESP8266 NodeMCU only writes the program into SRAM from USB. It is not the processor, and it is not tied to the CPU 5 V supply.
- Wired instructions: load, add, subtract, AND, OR, XOR, show the result on LEDs, halt, and no-operation. Jump and jump-if-zero are not wired.
- Check program the machine is built to run: load 5, add 7, show 12 (binary 0000 1100), then halt. Clock can free-run or single-step.

Anima — LLM interpretability meter (flagship)

Public · MIT · HF Spaces demo

- Forward hooks into Hugging Face causal LMs; probe heads read valence, arousal, and uncertainty per token.
- Guard recommends abstaining when the readout looks unreliable (HaluEval / TruthfulQA fixtures).
- Benchmarked on five open models: TinyLlama 1.1B scored 94/100 on the weighted validity rubric (60 = publication bar); GoEmotions valence Pearson $r \approx 0.19$.
- Technical: FastAPI + WebSocket stream, dashboard, public Spaces demo — meter, not a claim the model feels.

BumbleBee — Transformer-free CEC correlator (private)

Private repo · no Hugging Face transformer on the product path

- Closed-corpus ask / cite / refuse: GRU local encoder → SoftCorrelator (sparsemax) → CoverageGate → cite or refuse.

- Skills (chat / math / code) short-circuit via packs/tools; hard asks fall back to CEC. Not a frontier next-token LLM.
- Honesty: on the sealed pack BM25 still leads in-domain F1; skills wins ≠ core CEC quality. Patent: nothing filed.
- CLI: ingest, seal, train, calibrate, eval, ask, chat, dash. Footprint-first design, not dense-LLM parity.

Corvex — Multi-host campaign correlator (flagship)

Public · MIT · sealed synthetic eval · claim_allowed=false by default

- Fuses weak per-host detectors (lateral_auth, micro_exfil, recon_fanout) into ATT&CK-shaped campaign timelines.
- Observe-only Windows/macOS sensors; live containment locked off by default (human stays in control).
- Holds on synthetic fleets — not validated on real enterprise telemetry or a pure-benign baseline yet.
- Technical: HMAC-signed EventEnvelopes + transitive correlator merge across hosts.

GodFather — Multi-agent NSE paper desk (private)

Private repo · paper / session-sim only · not investment advice

- Scout / Risk / Session agents share one desk; Risk may cut size or force cash — never raise risk.
- Prior-session bulk/block BUY filings, open-range entry, same-day exits; CLAIMS.md honesty gates.
- Never sent a live broker order. Cut from the broader trade_bot lab.

Drift — Clinical rule-based health risk tracker

ACC/AHA · FINDRISC · safety-gated experimental ML

- Deterministic cardiovascular / diabetes risk calculators; ML path labeled experimental — not a medical device.
- Technical: Flask + SQLite; Fitbit / Google Fit optional sync.

GeoQuant — Algorithmic trading research platform

Walk-forward · cost-aware · honesty-first

- MLP signals, news sentiment, paper trading, retraining. Walk-forward backtests (2022 to 2025): v1 failed (Sharpe –0.47, kept public); v2 trend + vol target reached Sharpe 1.36.
- Backtests only; not investment advice. Technical: PyTorch + FastAPI with costs inside the optimizer.

Aerospace / simulation / drone arc (self-directed)

Simulation + hardware learning · vortex-tracker built during the IISc internship

- vortex-tracker (OpenCV diameter + speed) · Propeller-simulator (BEMT-style model, simplified) · Drone-Vortex-Ring-Simulation (reduced-order Helmholtz / Kelvin Γ; not CFD).
- NeuralVortex: early FNO-style volumetric surrogate (smoke-scale — not production accuracy).
- Homemade F450 quad + F550 hex builds; Pico FC learning firmware (props-off until tether tests pass).

Other supporting systems

- trade_bot: NSE bulk-deal ingest, scoring, WhatsApp alerts — lab that fed GodFather.
- text2sql-rag: Spider clean-room with schema linking, few-shot, sqlglot validation (evaluation pending).
- FleetControl: multi-host stub LLM memory pools + Anima probes.
- cursor-llm-council: multi-model Cursor council that resists yes-man answers.
- AI-BRAIN: voice-first daily OS (Whisper, local/cloud LLM, Qdrant RAG).
- Homelab: Raspberry Pi NAS + Pi-hole; Elevyx: wound-down real-estate lead recovery.

HARDWARE

- Discrete-logic CPU: built 8-bit 74HC breadboard computer (see project above). ESP8266 is the loader only.
- Multirotors: F450 quad + F550 hex (GPS / autonomous modes) — airframe, ESCs, FC tuning.
- Rovers: Bluetooth, line-follow, ultrasonic avoid; custom PCB plant-watering (soil moisture).
- Arduino / ESP32 toolkit including BLE macrodeck for workflow shortcuts.
- Neural Sync (in progress): low-cost 2-channel EEG research instrument — software-first, never mixes synthetic with live serial.

COMMUNITY · SPORTS

- Websites / volunteering: Project Thrive, Project Lighthouse, Arogi Foundation.
- Football: school team 4 years; state-level CBSE tournament; Goa Globe 2024, team 2nd place; ESSB / GameON.
- Past competitive skating (ForceOne Academy). Motorcycle track training (trained, not certified).
- Badminton, swimming, Model UN; two SWEA school trips (three days each); 50+ hours of community service.

Portfolio dossier (live widgets + GitHub index): https://siddarthb07.github.io/siddarthb/